

ORACLE DBA PRACTICE

An End-to-End Guide for Beginners

Practical, Step-by-Step Database Administration



Inside this sample

A practical starting point

Oracle DBA Practice: An End-to-End Guide for Beginners is a collection of practical database administration notes. This preview includes the contents and an excerpt on database startup stages and shutdown options.

What you can explore

The full notes cover installation, parameter files, tablespaces, users, backup and recovery, RMAN, networking, Data Pump, ASM, Data Guard and performance tuning. Full PDF/eBook price: INR 199.

About the examples

The source notes use older Oracle environments, including 11g and 12c. Commands, output and screenshots reflect those environments. This preview preserves the original excerpt rather than presenting it as a newly revised technical edition.

Reading and practice

Contents page references refer to the original notes, not this short preview. Use the examples in an isolated practice database. Startup and shutdown commands affect database availability; follow your organization's change process for live systems.

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Startup stages

Before using the oracle database we have to start the database instance.

To start the database instance we have to set instance name at os prompt.

```
$ export ORACLE_SID=<INSTANCE_NAME>
```

```
$ export ORACLE_SID=prod1
```

```
$ sqlplus / as sysdba
```

SQL*Plus: Release 11.2.0.3.0 Production on Thu Mar 24 21:15:22 2016

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Connected to an idle instance.

SQL> startup

ORACLE instance started.

Total System Global Area 418484224 bytes

Fixed Size 1345352 bytes

Variable Size 138414264 bytes

Database Buffers 272629760 bytes

Redo Buffers 6094848 bytes

Database mounted.

Database opened.

When Oracle is trying to open your database, it goes through three distinct stages..

These stages are:

Nomount Stage

During the nomount stage, Oracle first opens and reads the initialization parameter file (init.ora or spfile.ora) to see how the database is configured

SQL>startup nomount

ORACLE instance started.

Total System Global Area 418484224 bytes

Fixed Size 1345352 bytes

Variable Size 138414264 bytes

Database Buffers 272629760 bytes

Redo Buffers 6094848 bytes

Mount Stage

When the startup command enters the mount stage, it opens and reads the control file.

SQL> startup mount

ORACLE instance started.

Total System Global Area 418484224 bytes

Fixed Size 1345352 bytes

Variable Size 138414264 bytes

Database Buffers 272629760 bytes

Redo Buffers 6094848 bytes

Database mounted.

If you have already started the database instance with the startup nomount command, you might change it from the nomount to mount startup stage using the alter database command:

SQL> alter database mount;

Database altered.

The Open Stage

To open the database, you can just use the startup command as seen in this example

SQL> startup

ORACLE instance started.

Total System Global Area 418484224 bytes

Fixed Size 1345352 bytes

Variable Size 138414264 bytes

Database Buffers 272629760 bytes

Redo Buffers 6094848 bytes

Database mounted.

Database opened.

If the database is mounted, you can open it with the alter database open command as seen in this example:

SQL> alter database open;

Shutting Down a Database

To shut down a database and instance, you must first connect as SYSOPER or SYSDBA.

Shutting Down with various options

SQL>shutdown normal;

SQL> shutdown; Waits for all users sessions to be disconnected

SQL>shutdown immediate; Disconnects users sessions, rollbacks uncommitted data

SQL>shutdown transactional; Waits for all users sessions transactions to be completed

- in all above options, first checkpoint is performed

SQL>shutdown abort; crashes the instance, does not perform checkpoint, no rollback

- with above abort option, instance crash recovery is required on startup

Continue your learning

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Presented by S3 Academy

PDF/eBook: INR 199

This is a limited preview

This sample contains the contents and one selected practical topic. The remaining chapters are available in the full eBook.

Enquiries and portfolio

Visit MD Shahid Hussain's portfolio for book enquiries, remote Oracle Apps DBA support and training discussions:

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Thank you for reading

Consider whether the explanation style and practical examples suit your learning needs. Keep notes, practise carefully and build your understanding one topic at a time.